

Project Executive Summary

This project provides a comprehensive analysis of consumer shopping behavior to drive data-informed business strategies. By integrating Python for data processing, PostgreSQL for deep-dive querying, and Power BI for interactive visualization, the study evaluates the dynamics between customer demographics, subscription status, and product performance.

Key Findings

1. **Dominant Categories:**Clothing emerged as the premier category, leading both in total revenue generation and overall sales volume.
2. **Subscription Gap:**A substantial majority of the customer base consists of non-subscribers, representing a significant untapped opportunity for loyalty program expansion.
3. **Purchase Metrics:**The average transaction value stands at approximately \$59.76, providing a baseline for future up-selling strategies.
4. **Customer Satisfaction:**With an average review rating of 3.75, customer sentiment is generally positive, though there is room for optimization in product quality or service.
5. **Behavioral Variance:**Purchasing patterns shift noticeably across different product categories, necessitating tailored marketing approaches rather than a "one-size-fits-all" strategy.
6. **Revenue Concentration:**Financial returns are concentrated within a few core categories, allowing for more focused inventory management and promotional resource allocation.

Business Recommendations

- **Boost Subscription Growth:**Deploy targeted loyalty initiatives, such as exclusive subscriber-only discounts and reward point systems, to convert one-time shoppers into long-term subscribers.
- **Optimize High-Value Categories:**Double down on the Clothing segment by expanding product variety and increasing marketing spend where the Return on Investment (ROI) is highest.
- **Enhance Product Quality:**Conduct a deep dive into products with below-average ratings to address specific customer pain points and improve overall brand perception.
- **Precision Marketing:**Utilize customer segmentation based on demographics and past purchase history to deliver personalized promotions that increase conversion rates.
- **Agile Data Monitoring:**Maintain the Power BI dashboards for real-time tracking of KPIs and leverage predictive analytics to anticipate seasonal trends and manage inventory proactively.

Project Methodology

1. **Data Collection:**Aggregated consumer behavior data including demographics, purchase history, and feedback metrics.
2. **Data Preparation (Python):**Utilized Pandas for cleaning, handling missing values, and conducting Exploratory Data Analysis (EDA) to ensure data integrity.
3. **Database Analysis (PostgreSQL):**Engineered a relational database schema to perform complex queries on revenue distribution, payment preferences, and category performance.

4. **Insight Generation:**Evaluated the intersection of subscription status and spending habits to identify key business drivers.
5. **Visualization (Power BI):**Developed an interactive dashboard featuring KPI cards for average purchase values and ratings, alongside slicers for gender and category-specific deep dives.
6. **Strategic Documentation:**Synthesized technical findings into actionable business strategies and organized all assets—including Python scripts and SQL queries—into a structured repository for stakeholder review.